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Telecommunication:

They CAN SELL Only If It Complies With SAR TEST REQUIREMENTS

Telecommunication Engineering Centre and Department of Telecommunications have strict rules on SAR measurements, and it is mandatory for any mobile phone imported to India to be tested for SAR. Benoit Derat, PhD, chief executive officer and president, ART-Fi, and Rahul Gautam, head - market development, electromagnetic compatibility, Rohde & Schwarz, speaks with Sneha Ambastha, *Electronics Bazaar*, to share their views about the market requirement for SAR testing and its benefits

Q. What is SAR?

A. Specific absorption rate (SAR) is basically the amount of electromagnetic field produced by mobile phones. It is very important that we control it. Mobile manufacturers have to measure and show this value on their device in compliance with exposure limits to human body. They can only sell devices that comply with SAR test requirements.

Q. What is the difference between EMC/EMI testing and SAR testing?

A. Basic electromagnetic compatibility (EMC) or electromagnetic interference (EMI) testing is performed for interference between two machines. This is to ensure that radiation emitted by one machine does not disturb other machines. But SAR testing only looks at the effect of radiation on human beings.

Q. How has SAR testing evolved?

A. Development of wireless technology using traditional and slow methods has become difficult. This is because, with development of technologies like Long Term Evolution (LTE), the number of tests per device in order to perform SAR compliance is extremely large. Traditional compliance testing of a smartphone would

take around four to five weeks, but with the machine we developed, it could be finished in one week.

Q. How do newer SAR testing devices work?

A. Probes here are totally computerised and fixed, enclosed inside the head or body of the mannequins on the machine. The device itself does not move but is placed against the mannequin to simulate various conditions of mobile phone usage. The device is fixed in one position and probes inside the mannequin are activated electronically to perform a complete scan below the surface of the device. The scan captures the energy that is transferred from the device into the biological tissue-simulating material. Then, algorithms are applied to obtain the field everywhere you wish to have it.

Q. What are the specialties of this new SAR machine?

A. The former technology would measure only the magnitude of the electric field, but the new one measures both magnitude and phase. You can apply algorithms based on physical science to reconstruct the field in a complete volume from only a few points using equations. This is not possible with traditional systems.

Test time is also reduced due to multiple probes that are connected

electronically. These apply calculation algorithms in a fast computer to get the result delivered in seconds.

Q. What is the international situation on SAR testing?

A. In Europe, people more or less ignore it, although there are obligations for displaying it. World Health Organization supports the fact that there are some limits established by some expert committees that should not exceed in order to make sure that human health and safety is not at risk. That is why it is very important that we continue monitoring these types of values through equipment that can make those measurements.

Q. What is the critical SAR value?

A. Compliance limit varies across regions. India has chosen to align to the USA limit, which is one gram average SAR. This means SAR average on one gram of tissue is limited to 1.6W/kg. In other countries, limits are defined differently. For example, in Europe, ten grams SAR averaging is applied and the limit is 2W/kg.

Scientists want to take a cautious approach by putting a very low limit to make sure that we never get too close to the edge of the cliff. As of now, the limits prove to be safe by a large margin. **EFY**